Modeling with partial differential equations
Course overview

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Course content

Ordinary and partial differential equations are equations whose unknowns are functions, and which involve their derivatives

We will discuss four types of differential equations:

- Part 1: Ordinary differential equation: $\dot{x}(t) = f(x(t))$
- Part 2: Poisson equation: $-\Delta u = f$
- Part 3: Heat equation: $\partial_t u - \Delta u = f$
- Part 4: Transport phenomena: advection equation $\partial_t u + b \cdot \nabla u = f$, wave equation $\partial_{t,t} u - \Delta u = f$

We will study these equations using tools from *Classes préparatoires*, using only classical derivatives

Learning objectives

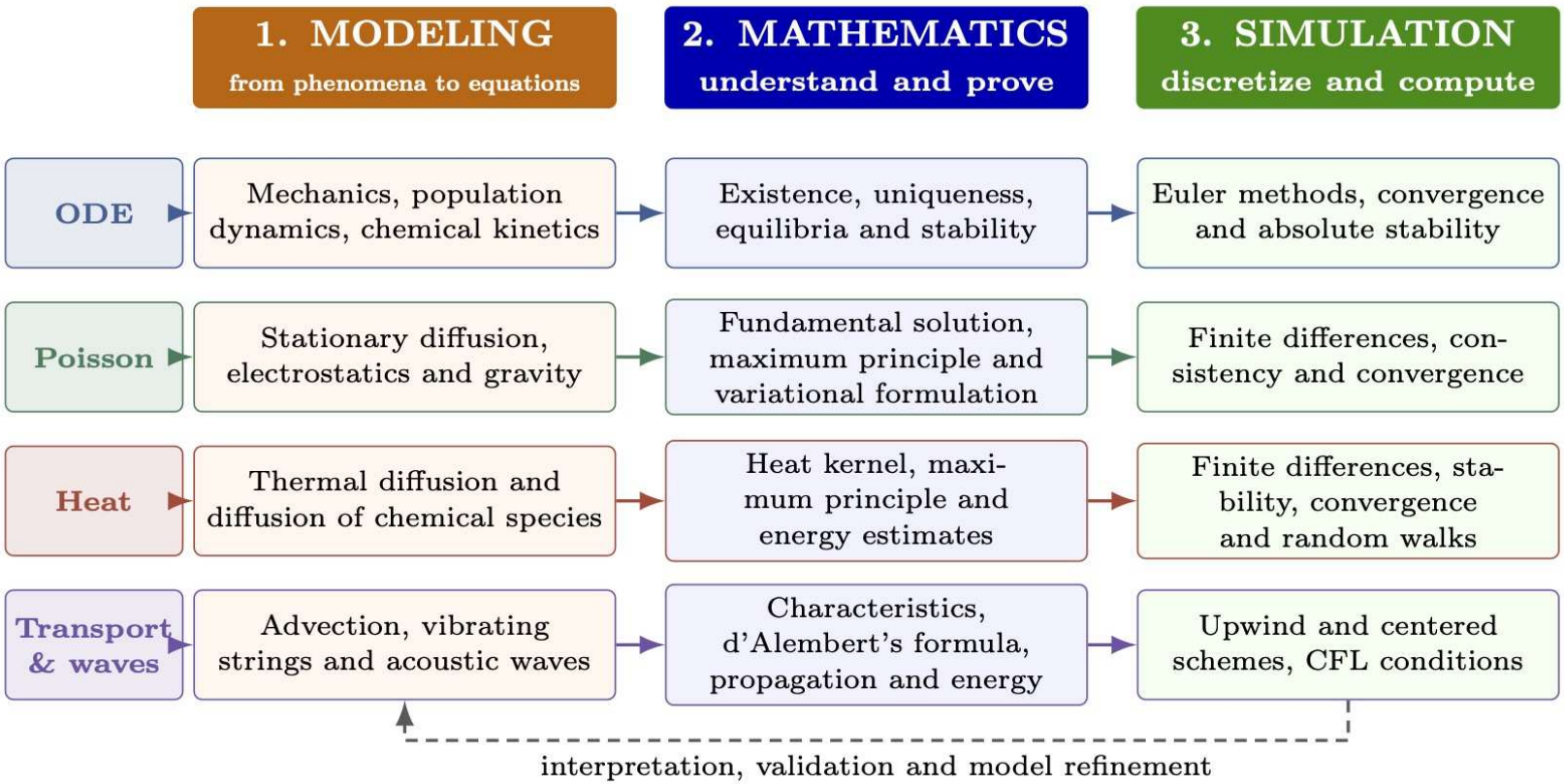
Objective:

- Study fundamental properties of differential equations and their discretizations
- Understand the interplay between modeling, mathematical analysis, and numerical simulations

These lectures are also introductory lectures for forthcoming mathematical courses:

- Analyse (première et deuxième partie)
- Equations aux dérivées partielles: approches variationnelles
- ... many lectures from the IMI department

Course structure



Prerequisites

We will need:

- Integration (multiple integrals, Lebesgue theorems for functions defined by integrals)
- some linear algebra (diagonalization, matrix exponential, matrix norm)
- Taylor expansions (on multivariate functions).

Pay close attention to Etienne Bernard's lectures (Pratique du calcul pour l'ingénieur)

Resources

Visit the website

`https://educnet.enpc.fr/course/view.php?id=2175`

You will find:

- The lecture notes (with a glossary of mathematical terms)
- The slides
- The exercise sheets (with corrections)
- Links to videos

Schedule

- Six lectures (3h): 1h lecture + 1h45 exercise session (bring a computer with internet connection):
  September 14, 21, 28, and October 5, 12, 19

ODEs		Poisson	Heat		Transport
Sept. 14	Sept. 21	Sept. 28	Oct. 5	Oct. 12	Oct. 19

- One pre-exam review session (3h): October 13
- Exam: November 2

Your work

- After each lesson:
  - Review the slides and exercises
  - Read the associated part in the lecture notes
  - Write a short summary (one hand-written double sided A4 sheet for each part)
- During the exam, you will be allowed to bring only the four hand-written double sided A4 sheets associated with each part of the lectures. Only the notions seen during the lectures (amphi) and the exercise sessions are necessary to pass the exam